

Bugs found: 3

1. Use-after-free: pointer `p` is dereferenced (`*p = 42`) immediately after being freed with `free(p)` in `process()`.
2. Double free: `ptr` is freed inside `process()` via `free(p)`, then freed again in `main()` with `free(ptr)`, causing undefined behavior.
3. Integer overflow: the loop accumulates a sum up to $(99999 * 100000 / 2 = 4,999,950,000)$, which exceeds the range of a 32-bit signed int, causing overflow.